AI Alliance Drone Challenge Rules

Competition Walkthrough- This game involves strong cooperation between the Autonomous HULA drone and the manual HULA drone.



2. Aircraft requirements

Model	HULA Quadcopter
Wheelbase	120~140mm
Motor type	Coreless motor
Takeoff weight	≤120g(Includes protective cover and battery)
	rotection design At least a semi-enveloping structural protective
Battery life	>7 minute
Battery Type	lithium battery
Battery parameters	1S, rated voltage3.8V,capacity≤1300mAh
programming language	Graphical programming or Python programming

Programming software	Yes, support flight simulation preview
aircraft lights	Yes, programmable
Visual Identity	Yes, programmable, able to identify identification tags in the field
Four-way obstacle avoidance	Yes, programmable
PTZ	Yes, supports pitch adjustment and is programmable
Laser transmitting and receiving	Yes, programmable

Positioning plan	TOF Positioning, optical flow positioning
programming equipment	Computer, mobile device, Ipad

Team members

Up to 3 team members from the same school

1 Student mentor is allowed to assist each team in the 1 minute preparation phase.

Student mentor (NEW)

Criteria for student mentor

1) Student mentor must not be taking part in the Air Land Conquest 2026 (any category).
2) Student mentor must be an alumni or existing student of the same school with the team (to be verified by the teacher).
3) Student mentor must be an existing student in any educational institution and below the age of 24 years old. (verified by valid student/ matriculation pass)
4) Student mentor must not be a paid external trainer , engaged by the school (as verified by the teacher in charge)

Student mentors will receive certificates of recognition and medals of recognition .Medals of varying recognition depends on the quantitative performance of the mentored team, as well as qualitative performance such as teamwork as observed by judges during their competition round.

Changes in 2026

1) Door mission is removed

2) Manual and autonomous drone cannot deviate more than 50 cm from the line

Competition Walkthrough
Surprise component may be revealed on competition day.

https://youtu.be/bFNM23MIikU

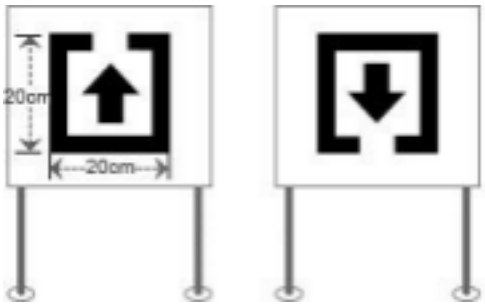
Competition site – 6m by 5m
Duration of game : Up to 6 minutes
Preparation time - 1 minute

To start : Autonomous drone must take off first , emit laser beam, as a cue for the manual drone to start its mission.

Timing for racer drone - starts after racer manual drone takes off . Racer manual drone can only take off after autonomous drone take off and emit laser beam.

If Autonomous drone do not take off and emit laser beam, but racer manual drone takes off. Timing for racer manual drone will be invalid.

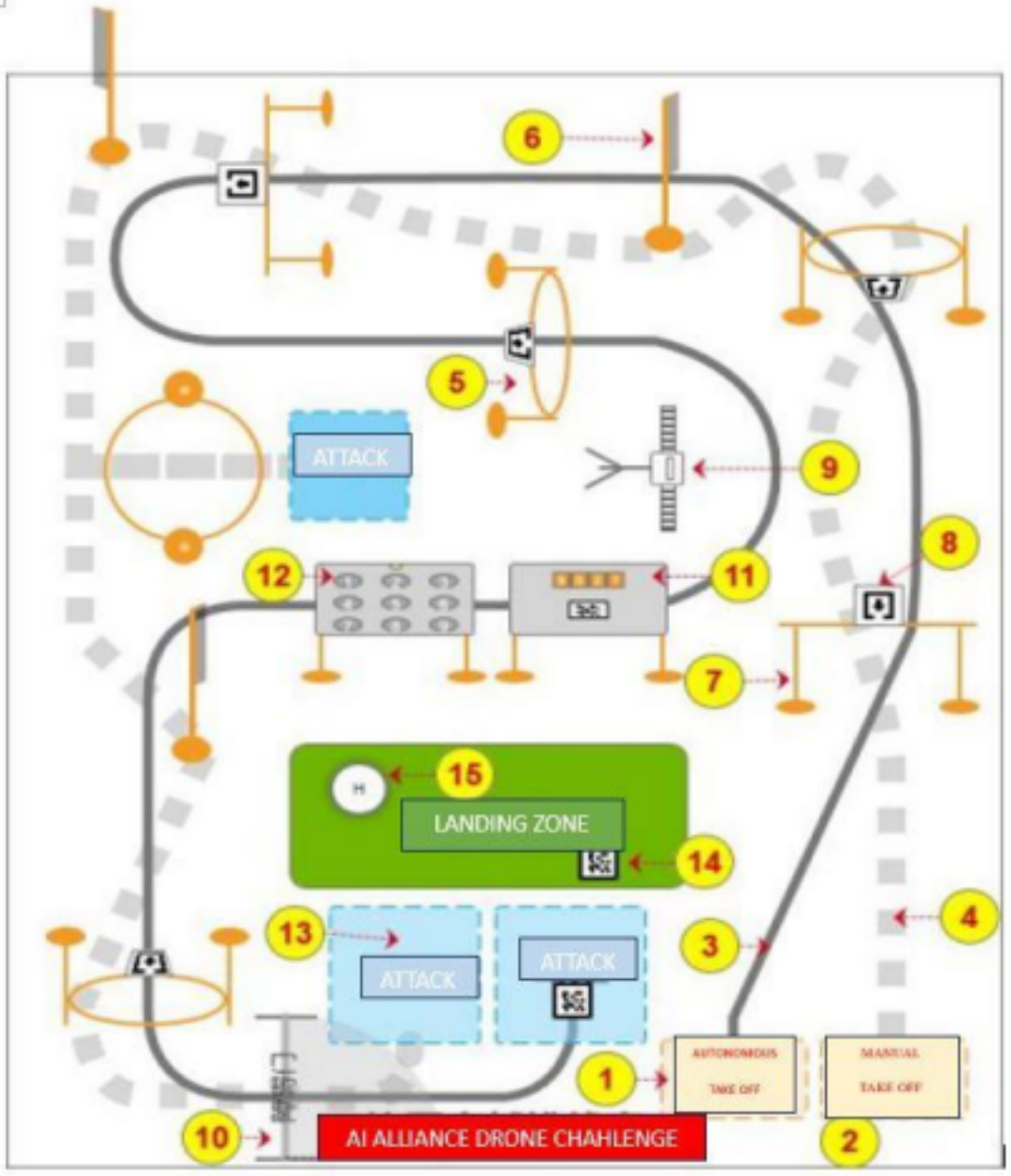
1. Autonomous drone
   Take off at " Autonomous take off zone" pad (at 1)and follow thick line as flight path, identify different directions and move through different obstacles to reach to the attack zone.



Note: Directional label reminder cards will be placed around the obstacles in

different forms such as standing, mounting, and pasting.

   There is a " door " (at 10) and door will open , only when manual drone manage to fire accurately at moving target board (at 9) .DOOR mission is removed
2. Manual drone
   Take off at " Manual take off zone" pad (at 2) and follow dotted line as flight path and move through different obstacles to reach to the 1st attack zone (at 9) to fire accurately at moving target board to open the " door" (at 10)
3. AI Alliance mission
   Manual drone must shoot moving target board accurately (at 9) to open door (at 10)in order to enable both manual and autonomous drones to move through.
   Manual drone need only to emit laser to shoot target board. Door mission is removed.
   (At 13) , the autonomous drone must shoot the QR code (at 11) to activate 4 numbers. The manual drone must shoot the 4 numbers (at 12). Once all 4 numbers are shot, both manual and autonomous drone must land at landing zone (at 15 and 14 respectively).



[1] Line patrol take-off area [2] Crossing take-off area [3] Line patrol track [4] Crossing track [5] Ring [6] Knife flag [7] Arch [8] Direction label [9] Target target [10] Collaborative obstacle door [11] Password acquisition point [12] Password wall [13] Strike area [14] Line patrol landing point [15] Crossing landing point [16] Distance ruler

3. Performance evaluation

1. Time rules

1) Each team is given 1 minute preparation time

2) Each game is limited to 6 minutes. When the game starts, the referee will issue an instruction and start timing. The referee will record the total time taken by the drone to complete the competition task.

3) The conditions for drones to complete the competition mission are: both drones

land in the designated landing area. At the end of the game, the referee immediately stops the clock.

3) If the 6 minutes of the competition end and the drone has not completed the competition task, the competition will end immediately, the competition results will be tallied, and the contestants must immediately control the drone to land on the spot.

4) Scoring details

Drones will receive corresponding bonus and subtraction points rewards based on the completion of relevant tasks. The specific details are as follows:

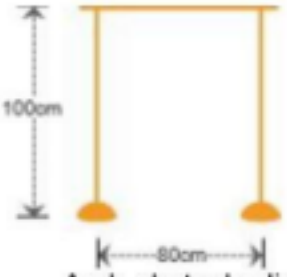
S/N	Penalty item	marks
1	Completion points	150
2	Autonomous drone light up when not required	-1 per incident
3	Crash when going through obstacle	-1 per incident
4	Did not go through the obstacle	-5 per incident
5	Did not shoot in the attack zone	-5
6	Slight Deviation from flight path	-5
7	Did not open the door but instead go through another route	-10
8	Did not complete the shooting of numbers	-10
9	Failed Landing	-10
10	Autonomous drone does not follow manual drone laser cue to take off	-40
11	Autonomous drone does not follow directions to fly	-60
12	Unlawful behaviour such as cheating	Disqualification

Obstacle dimensions

The ground material of the venue is a special customized carpet, with visual characteristic patterns printed on it for drone positioning. 6. Site obstacles: There are three types of site obstacles arranged in the venue, namely arches, sword flags, and rings. Among them, three types of obstacles, namely arches, sword flags and rings, will be arranged in the crossing mission track, and two types of obstacles, arches and rings, will be arranged in the line-following mission track.

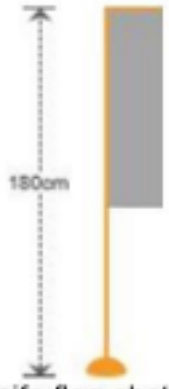
1) Arch: The arch is a rectangular arch with a height of 100cm and a width of 80cm. Crossing mission crossing rules: the drone must pass through the inside of the arch to be considered a successful challenge; line patrol mission crossing rules: crossing the arch obstacle

When the drone passes through the arch, the LED lights of the drone need to turn on yellow at the same time; after the drone passes through the arch, the LED lights need to be turned off.



Arch obstacle diagram

2) The knife flag: The knife flag is a vertical flag with a height of 180cm. Crossing mission rules: The drone must pass through the knife flag to be considered a successful challenge.



Knife flag obstacle diagram

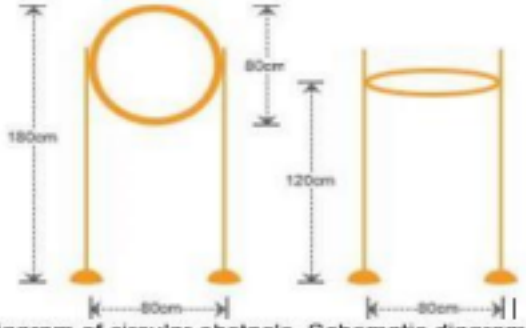
3) Ring:

a. There are two types of circular obstacles: circular obstacles and vertical circular obstacles.

b. The height of the ring obstacle is 180cm, and the diameter of the inner circle is 80cm. The vertical ring obstacle is 120cm high and the inner ring diameter is 80cm.

c. Crossing mission crossing rules: The drone must pass through the inner circle of the ring to be considered a successful challenge. The vertical ring obstacle needs to be crossed twice from top to bottom/bottom to top, and the crossing width must be at least 50cm.

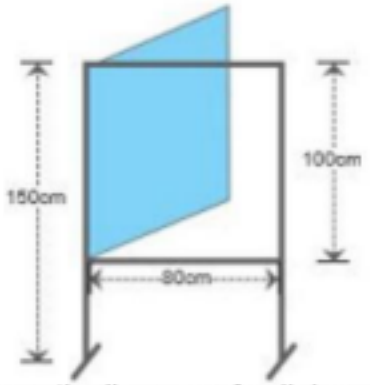
d. Line patrol task crossing rules: When crossing the ring obstacle, the drone's LED light needs to be turned on green at the same time. After the drone passes through the ring, the LED light needs to be turned off.



Schematic diagram of circular obstacle. Schematic diagram of vertical circular obstacle.

4) Collaborative obstacles: There are collaborative obstacles arranged in the competition, which consist of target targets, collaborative obstacle doors, password acquisition points and password walls. Two drones that need to perform crossing tasks and line patrol tasks perform different tasks respectively. Assist the other party in completing a task.

6) Collaborative barrier door: The collaborative barrier door is 150cm high and 80cm wide, and consists of a door frame and a movable door. The cooperative obstacle door is placed on the track of the line patrol mission. When the drone that needs to perform the crossing mission emits a laser and hits the target, the movable door will open. After the cooperative obstacle door is opened for a period of time, it will automatically close. To successfully pass, you need to hit the target again to open the cooperative obstacle door.



Schematic diagram of collaborative barrier door

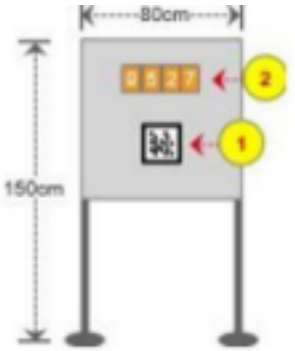
7) Password acquisition point & password wall a. The password acquisition point is

150cm high, with a standing sign 80cm wide and 80cm high above it. There is a QR code label (number 0) and a 4-digit customs clearance password display on the wall.

b. The password acquisition point is placed at the end of the line patrol track. When the drone reaches the end of the track, it needs to aim at the QR code and fire a laser to hit the QR code, and then the 4-digit code will be displayed in the password display area.

Password.

c. The pass code will be automatically and randomly refreshed for each game.



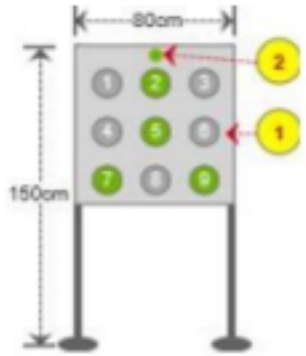
Password acquisition point diagram
[1] QR code label (No. 0) [2] Password display area

d. The password wall is 150cm high, with a standing sign 80cm wide and 80cm high above it. A password wall composed of lighted keyboards with passwords No. 1-9 on the wall.

e. The password wall is placed at the end of the track across the mission area. The drone needs to emit an infrared laser in the strike area and successfully hit the correct 4-digit password (in no particular order) before issuing instructions.When the light comes on, it is considered a success.

f. When the drone hits the correct digital code, the digital code lights up green. When the drone hits the wrong password, all lights will go out and you need to shoot again and enter it again.

Enter the 4-digit pass code.



Password wall diagram

[1] Password light keyboard [2] Indicator light

8) Landing zone: After completing the crossing mission and line patrol mission, the drone must land at the designated landing location in the landing zone to be deemed to have finally completed the event mission.

9) Line patrol landing point

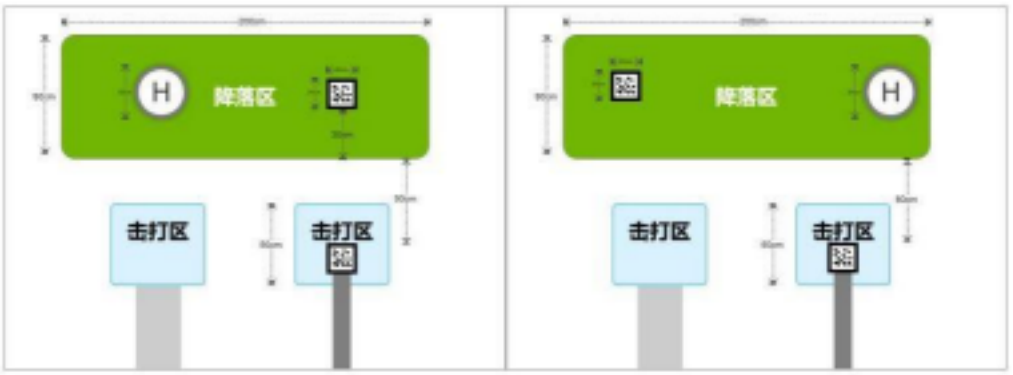
a. The drone performing the line patrol mission must accurately land at the QR code (No. 0) in the landing area. The QR code (No. 0) is a rectangular QR code label card with a length of 20cm and a width of 20cm.

b. At the end of the line patrol task (on the ground), there is a QR code (No. 1) at the end of the line patrol track to help determine whether the player has reached the end of the track.

c. There is a distance ruler between the line patrol landing point of the beginner group and the end of the line patrol track, and the position is fixed on the extended line directly in front of the line patrol track. d. The line patrol landing point of the high-level group will be randomly placed at any location in the landing area, and there will be no distance ruler.

10. Crossing the landing point

a. The drone performing the crossing mission must accurately land at the landing point "H" in the landing area. The "H" landing point is a 30cm long and 30cm wide round label card.



attack

Penalty for serious violation

The penalty system for the Challenge is as follows:

Penalty	Cause
Verbal warning	If a player violates the rules, the referee will give a verbal warning.
Reset of results to zero	If there is a violation by the participating team member or drone, the drone must be restarted at the starting point, and the previously obtained results must be cleared.
Disqualification	If there is a serious violation of the competition by a team member or drone, the referee will disqualify the participating team from the competition.

Rule	Penalty
1. Once competition has started, players are not allowed to move the drone	Results will be reset to zero. Return back the drone to original position. Timer will still continue
2. Players are not allowed to take off the drone before the referee signal start.	Disqualification
3. Once competition has started,without judges' permission, players are not allowed to alter their code or change their drone	Disqualification
4. Teams are not allowed to share codes	Disqualification

Drone movements during the competition

Rule	Penalty
Drones are to follow their respective flight path.	Results will be reset to zero if the drone deviate from its original flight path for more than 5 seconds.
Manual drone is only to be manually operated by the manual drone pilot and not to be interfered via any other means by external party	disqualification
No short cut allowed when following flight path	disqualification